Exercise - SendGrid Connector

In this exercise, you will modify the *Entertain Me* process so that the joke is sent via email.

The *Send Joke* task will use the **SendGrid Connector** to send the joke to an email address of your choice.

Create a SendGrid Account

To use the **SendGrid Connector**, create a free account in SendGrid if you do not have one yet:

1. Go to https://signup.sendgrid.com/.
2. Set up the account with your email and choose a password.
3. Click **Create Account**.
4. Provide further information required by SendGrid.
5. Click **Get Started**.

Create a sender identity

Before sending your first email, you'll need to create a sender identity and verify it.

1. Click **Settings > Sender Authentication** or click here.
2. Choose **Verify a Single Sender** for demo purposes (or alternatively **Authenticate Your Domain** for a production setup.)
3. Provide the details requested by SendGrid in the form and click **Create**.
4. Go to your email inbox and open the email sent to you by SendGrid.
5. Click **Verify Single Sender**.

Create an API key

To create an API key in SendGrid, take the following steps:

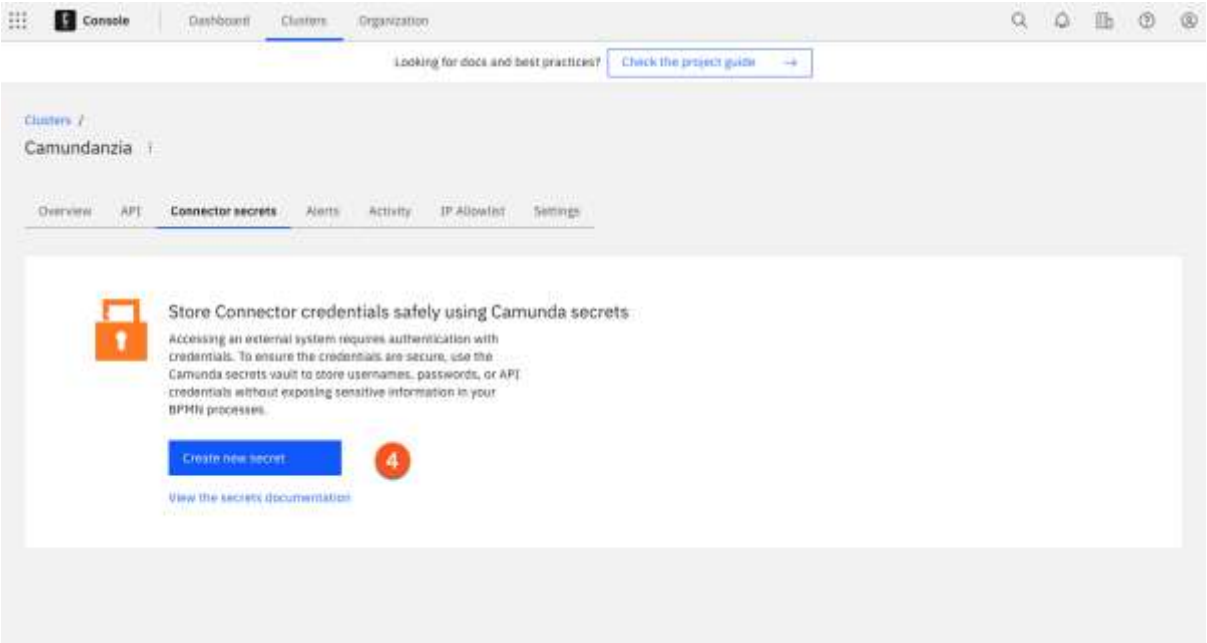
1. Log in to your new account.
2. Go to **Settings**.
3. Click **API Keys > Create API Key**.
4. Give your key a name (i.e. `my-camunda-connector-key`).
5. Click **Create Key**.
6. Copy the **API Key** and move on to the next step for creating a Connector secret.

Create a Secret

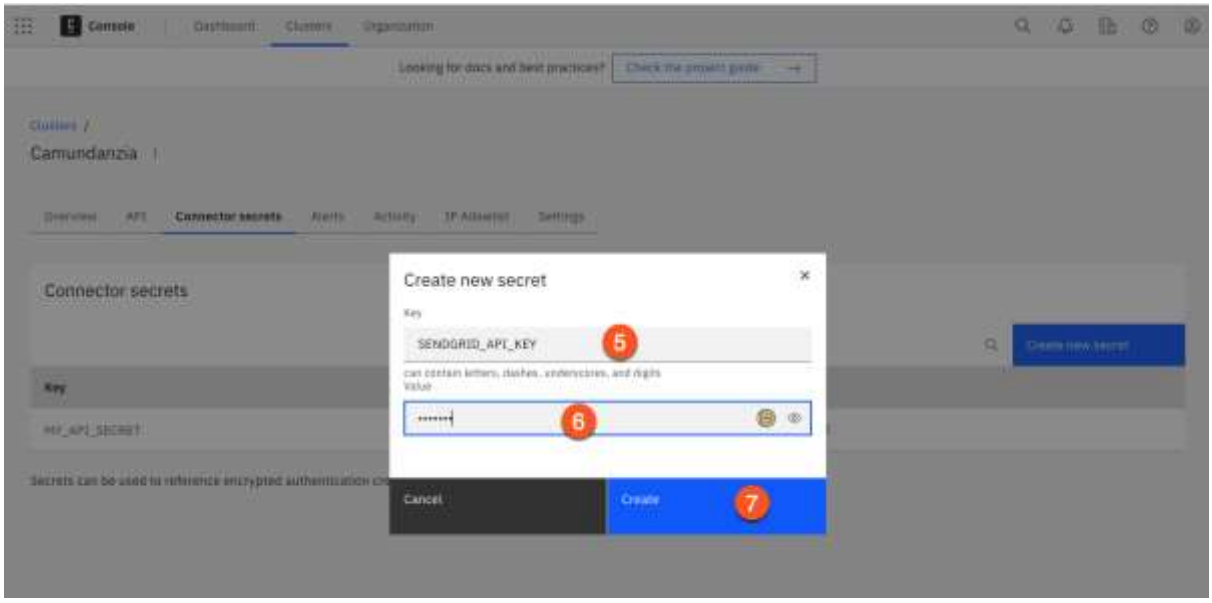
We first need to create a **Secret** to securely store our **SendGrid API Key** by following the steps below:

1. Navigate to the **Console**
2. Select the **Cluster** that your *Entertain Me* process has been deployed to
3. Click the **Connector Secrets** tab

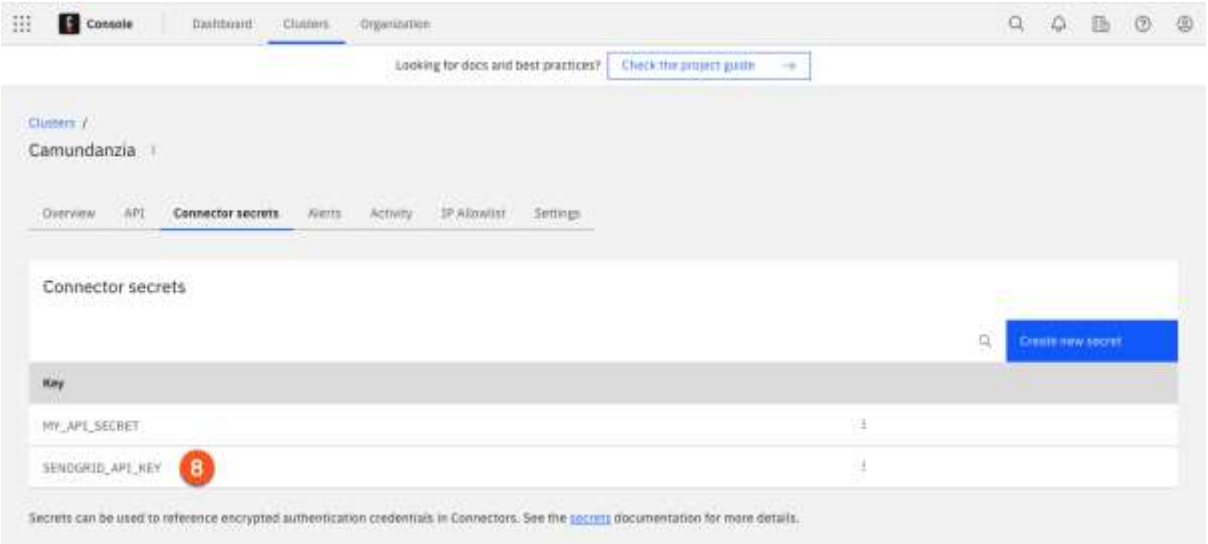
4. Click the **Create New Secret** button



5. In the **Create New Secret** dialog box, enter `SENDGRID_API_KEY` for the **Key**
6. Enter the API Key that you obtained earlier from SendGrid for the **Value**
7. Click **Create** to create the API Key



8. Your newly created **Secret** should now be visible in the **Connector Secrets** tab



Configure the Send Joke Task

You will now need to configure the *Send Joke* Task to use the **SendGrid Connector**.

1. Launch the **Web Modeler**
2. Add a new **Task** between *Generate joke* and *Joke received* with the following properties

Name	Value
Name	Send joke
ID	Task_SendJoke

3. Select the *Send Joke* task, click on the **+Select** button in the **Template** section of the properties panel and choose **SendGrid Outbound Connector** from the menu that appears
4. Configure the **Authentication** properties as follows:

Name	Value
SendGrid API Key	{{secrets.SENDGRID_API_KEY}}

Authentication

We are using the **Secret** that was previously configured in the cluster instead of hard coding the API Key.

Deprecated Syntax

The `secrets.*` syntax has been deprecated. Instead, use `{{secrets.*}}`

5. Configure the **Sender** properties as follows:

Name	Value
Name	Your Name
Email Address	Your Email Address

Sender

For the purposes of this course, you should configure yourself as the **Sender** of the email.

6. Configure the **Receiver** properties as follows:

Name	Value
Name	Your Name
Email Address	Your Email Address

Receiver

For the purposes of this course, you should configure yourself as the **Receiver** of the email.

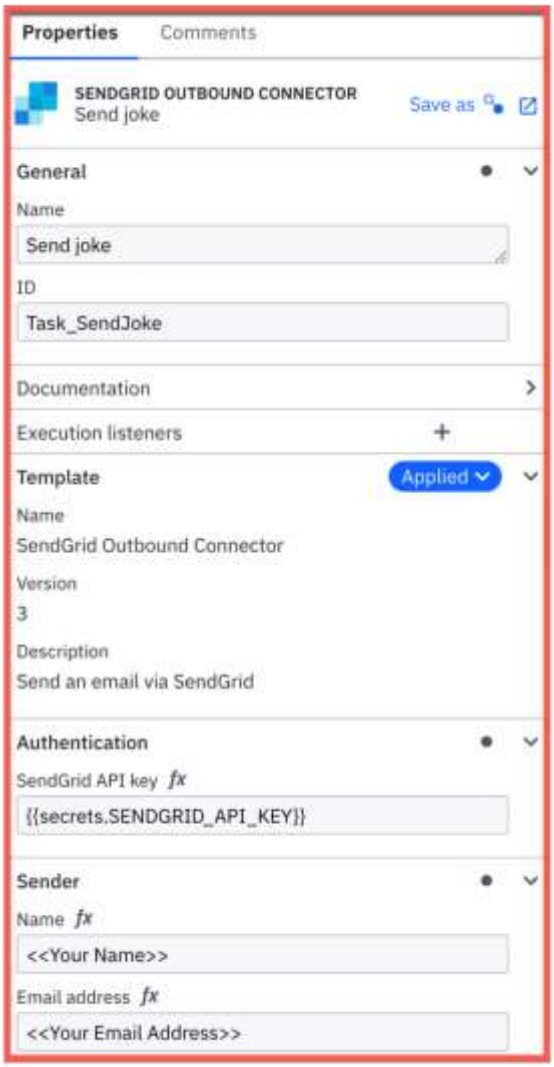
7. Configure the **Compose Email** properties as follows:

Name	Value
Compose Email	Simple (no dynamic template)
Subject	Your joke is here!
Content Type	text/plain
Body	setup + " " + punchline

Email Templates

We have used a simple FEEL expression here to deliver the `setup` and `punchline`. You can also create more complex emails using Email Templates.

8. The properties of the *Send Joke* task should look like this now:



Test the Process

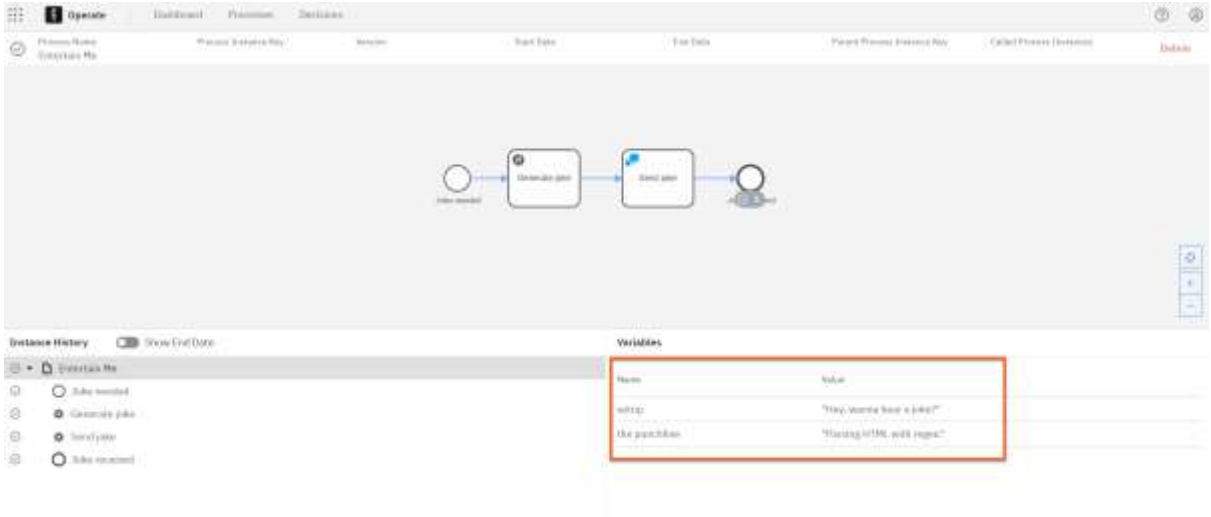
You can now test the **SendGrid Connector** by deploying your process and starting an instance.

1. Deploy the updated version of the process to your cluster by clicking the **Deploy** button
2. Click the **Run** button to start a new instance of the process
3. Launch **Operate** and select the *Entertain Me* process from **Process Instances by Name**

Operate

You can switch to **Operate** using the **App Switcher** in the top left corner of the screen. Alternatively, you can click on the **Instance Started** pop-up at the bottom of the screen which is displayed for a few seconds after starting your instance.

4. Click on the **Process Instance Key** for the process that you have just started; it should have completed immediately
5. You should be able to see that your process instance has completed successfully and the `setup` and `punchline` variables have been populated as **Process Variables**



6. Finally, you should also receive an email from yourself with the joke returned by the API

Configured the SendGrid Connector

Congratulations! You have created a process which uses the **REST Connector** to obtain data from an external service and then uses the **SendGrid Connector** to email that data.

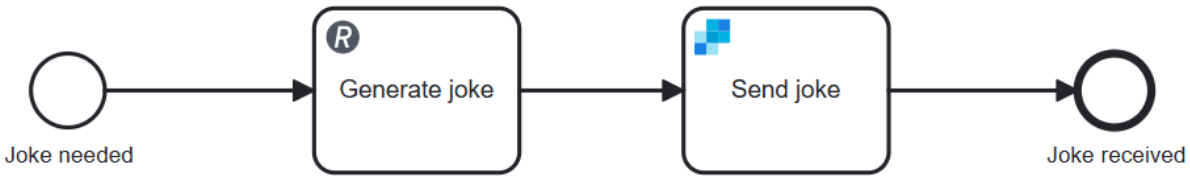
Solution - SendGrid Connector

If you encountered any problems during this course or were unable to complete the process configuration, you can compare your solution with our example solution below.

GitHub Repository

Our version of the **Process Definition** including the **REST Connector** and **SendGrid Connector** configuration can be found in the GitHub Repository below:

- Solution - Entertain Me

Process Diagram

Our version of the **Process Diagram** can be found below: